



NATIONAL UNIVERSITY
of Computer & Emerging Sciences

Subject: Digital Signal Processing

Instructor: Dr. Muhammad Akmal

Email: m.akmal@nu.edu.pk

Q2.7 (a) Is the following system stable or not?

$$y(n) = \cos[x(n)]$$

Solution:

As we know that

$$-1 \leq \cos[x(n)] \leq 1$$

for all values of $x(n)$

Hence, the system is stable
irrespective of the $x(n)$.



Scanned with CamScanner

2.6 CORRELATION OF DISCRETE-TIME SIGNALS

A mathematical operation that closely resembles convolution is correlation. Just as in the case of convolution, two signal sequences are involved in correlation. In contrast to convolution, however, our objective in computing the correlation between the two signals is to measure the degree to which the two signals are similar and thus to extract some information that depends to a large extent on the application. Correlation of signals is often encountered in radar, sonar, digital communications, geology, and other areas in science and engineering.

To be specific, let us suppose that we have two signal sequences $x(n)$ and $y(n)$ that we wish to compare. In radar and active sonar applications, $x(n)$ can represent the sampled version of the transmitted signal and $y(n)$ can represent the sampled version of the received signal at the output of the analog-to-digital (A/D) converter. If a target is present in the space being searched by the radar or sonar, the received signal $y(n)$ consists of a delayed version of the transmitted signal, reflected from the target, and corrupted by additive noise. Figure 2.38 depicts the radar signal reception problem.

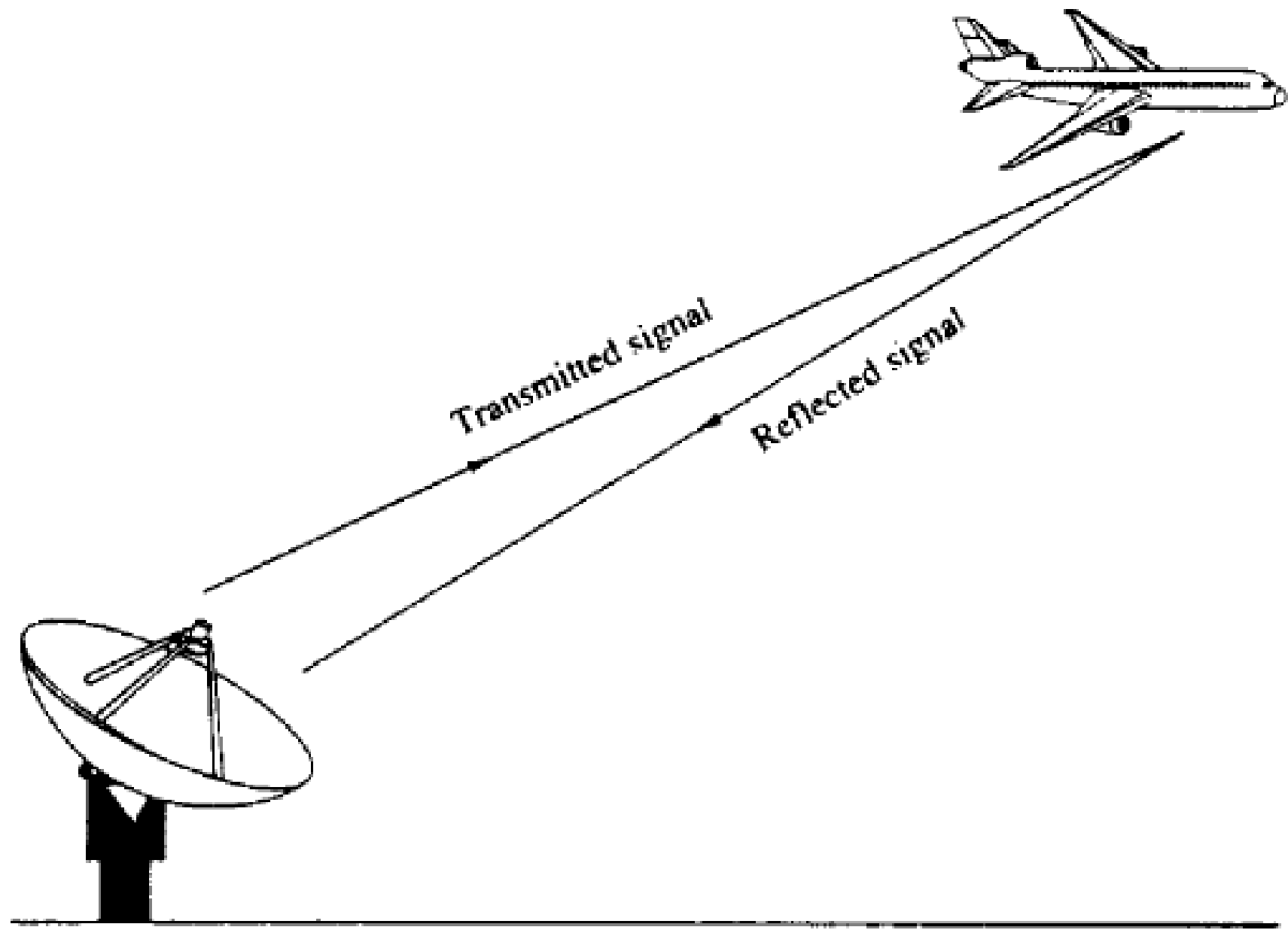


Figure 2.38 Radar target detection.

We can represent the received signal sequence as

$$y(n) = \alpha x(n - D) + w(n) \quad (2.6.1)$$

where α is some attenuation factor representing the signal loss involved in the round-trip transmission of the signal $x(n)$, D is the round-trip delay, which is assumed to be an integer multiple of the sampling interval, and $w(n)$ represents the additive noise that is picked up by the antenna and any noise generated by the electronic components and amplifiers contained in the front end of the receiver. On the other hand, if there is no target in the space searched by the radar and sonar, the received signal $y(n)$ consists of noise alone.

Having the two signal sequences, $x(n)$, which is called the reference signal or transmitted signal, and $y(n)$, the received signal, the problem in radar and sonar detection is to compare $y(n)$ and $x(n)$ to determine if a target is present and, if so, to determine the time delay D and compute the distance to the target. In practice, the signal $x(n - D)$ is heavily corrupted by the additive noise to the point where a visual inspection of $y(n)$ does not reveal the presence or absence of the desired signal reflected from the target. Correlation provides us with a means for extracting this important information from $y(n)$.

L is some integer that denotes the number of samples in each of the two sequences. Very often, $x_1(n)$ is selected to be the negative of $x_0(n)$. The signal received by the intended receiver may be represented as

$$y(n) = x_i(n) + w(n) \quad i = 0, 1 \quad 0 \leq n \leq L - 1 \quad (2.6.2)$$

where now the uncertainty is whether $x_0(n)$ or $x_1(n)$ is the signal component in $y(n)$, and $w(n)$ represents the additive noise

2.6.1 Crosscorrelation and Autocorrelation Sequences

Suppose that we have two real signal sequences $x(n)$ and $y(n)$ each of which has finite energy. The *crosscorrelation* of $x(n)$ and $y(n)$ is a sequence $r_{xy}(l)$, which is defined as

$$r_{xy}(l) = \sum_{n=-\infty}^{\infty} x(n)y(n-l) \quad l = 0, \pm 1, \pm 2, \dots \quad (2.6.3)$$

The index l is the (time) shift (or *lag*) parameter and the subscripts xy on the cross-correlation sequence $r_{xy}(l)$ indicate the sequences being correlated. The order of the subscripts, with x preceding y , indicates the direction in which one sequence is shifted, relative to the other. To elaborate, in (2.6.3), the sequence $x(n)$ is left unshifted and $y(n)$ is shifted by l units in time, to the right for l positive and to the left for l negative.

Example 2.6.1

Determine the crosscorrelation sequence $r_{xy}(l)$ of the sequences

$$x(n) = \{\dots, 0, 0, 2, -1, 3, 7, 1, 2, -3, 0, 0, \dots\}$$

$\uparrow$

$$y(n) = \{\dots, 0, 0, 1, -1, 2, -2, 4, 1, -2, 5, 0, 0, \dots\}$$

$\uparrow$

Solution Let us use the definition in (2.6.3) to compute $r_{xy}(l)$. For $l = 0$ we have

$$r_{xy}(0) = \sum_{n=-\infty}^{\infty} x(n)y(n)$$

The product sequence $v_0(n) = x(n)y(n)$ is

$$v_0(n) = \{\dots, 0, 0, 2, 1, 6, -14, 4, 2, 6, 0, 0, \dots\}$$

$\uparrow$

and hence the sum over all values of n is

$$r_{xy}(0) = 7$$

For $l > 0$, we simply shift $y(n)$ to the right relative to $x(n)$ by l units, compute the product sequence $v_l(n) = x(n)y(n-l)$, and finally, sum over all values of the product sequence. Thus we obtain

$$\begin{aligned} r_{xy}(1) &= 13, & r_{xy}(2) &= -18, & r_{xy}(3) &= 16, & r_{xy}(4) &= -7 \\ r_{xy}(5) &= 5, & r_{xy}(6) &= -3, & r_{xy}(l) &= 0, & l &\geq 7 \end{aligned}$$

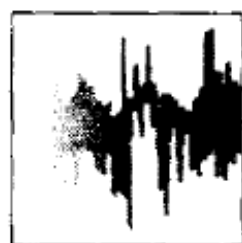
For $l < 0$, we shift $y(n)$ to the left relative to $x(n)$ by l units, compute the product sequence $v_l(n) = x(n)y(n-l)$, and sum over all values of the product sequence. Thus we obtain the values of the crosscorrelation sequence

$$\begin{aligned} r_{xy}(-1) &= 0, & r_{xy}(-2) &= 33, & r_{xy}(-3) &= -14, & r_{xy}(-4) &= 36 \\ r_{xy}(-5) &= 19, & r_{xy}(-6) &= -9, & r_{xy}(-7) &= 10, & r_{xy}(l) &= 0, \quad l \leq -8 \end{aligned}$$

Therefore, the crosscorrelation sequence of $x(n)$ and $y(n)$ is

$$r_{xy}(l) = \{10, -9, 19, 36, -14, 33, 0, 7, 13, -18, 16, -7, 5, -3\}$$





4

Frequency Analysis of Signals and Systems

The Fourier transform is one of several mathematical tools that is useful in the analysis and design of LTI systems. Another is the Fourier series. These signal representations basically involve the decomposition of the signals in terms of sinusoidal (or complex exponential) components. With such a decomposition, a signal is said to be represented in the *frequency domain*.

As we shall demonstrate, most signals of practical interest can be decomposed into a sum of sinusoidal signal components. For the class of periodic signals, such a decomposition is called a *Fourier series*. For the class of finite energy signals, the decomposition is called the *Fourier transform*.

4.2 FREQUENCY ANALYSIS OF DISCRETE-TIME SIGNALS

4.2.1 The Fourier Series for Discrete-Time Periodic Signals

Suppose that we are given a periodic sequence $x(n)$ with period N , that is, $x(n) = x(n + N)$ for all n . The Fourier series representation for $x(n)$ consists of N harmonically related exponential functions

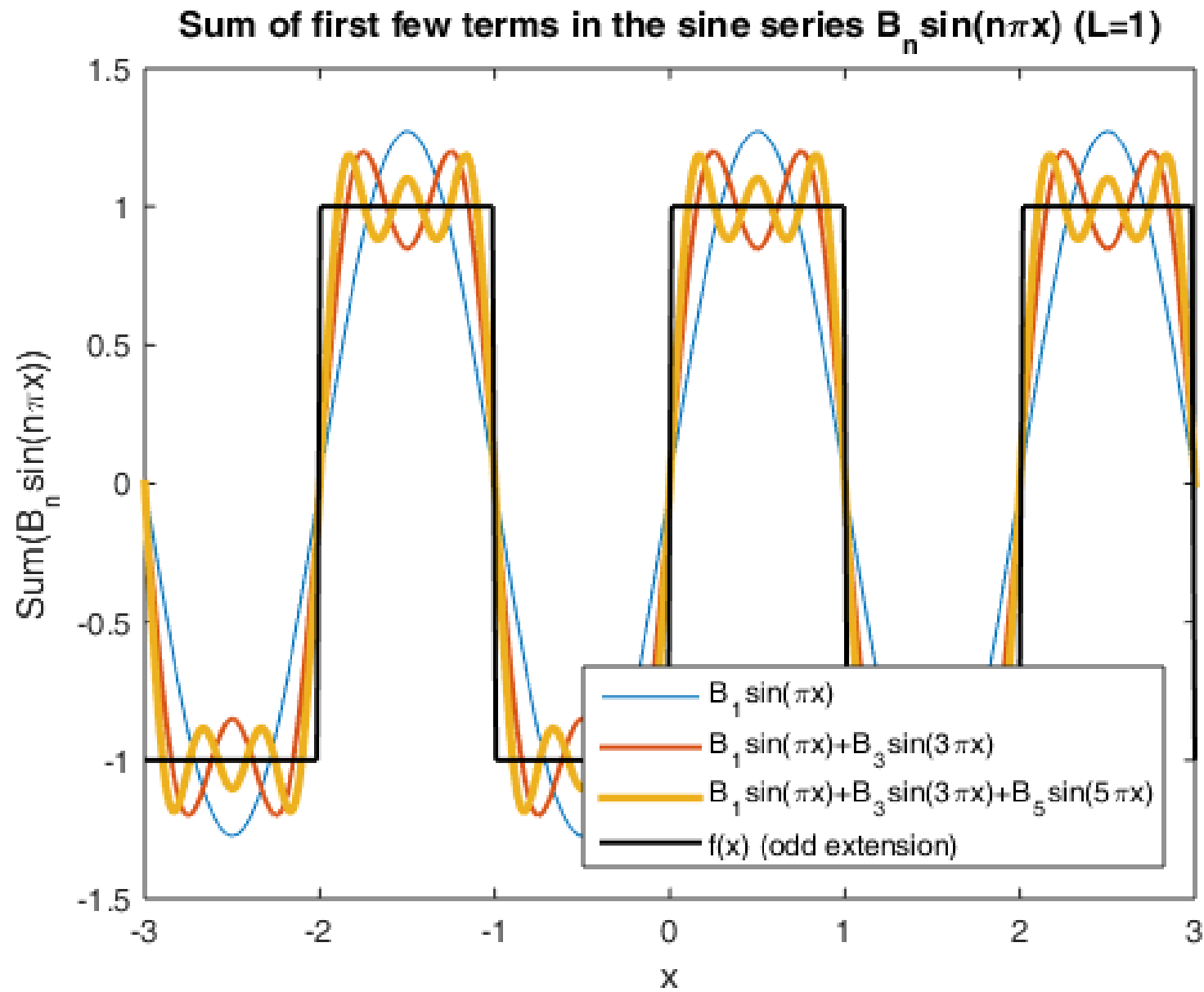
$$e^{j2\pi kn/N} \quad k = 0, 1, \dots, N - 1$$

and is expressed as

$$x(n) = \sum_{k=0}^{N-1} c_k e^{j2\pi kn/N} \quad (4.2.1)$$

where the $\{c_k\}$ are the coefficients in the series representation.

Any periodic
sequence $x(n)$ can
be represented by
4.2.1



To derive the expression for the Fourier coefficients, we use the following formula:

$$\sum_{n=0}^{N-1} e^{j2\pi kn/N} = \begin{cases} N, & k = 0, \pm N, \pm 2N, \dots \\ 0, & \text{otherwise} \end{cases} \quad (4.2.2)$$



Let's first understand it in the next slide

$$\sum_{n=0}^{N-1} e^{j2\pi kn/N} = \begin{cases} N & k=0, \pm N, \pm 2N, \dots \\ 0 & \text{otherwise} \end{cases}$$

Let $k=0$:

$$\sum_{n=0}^{N-1} e^0 = N$$

Let $k=N=3$:

$$\begin{aligned} \sum_{n=0}^2 e^{j2\pi 3n/3} &= \sum_{n=0}^2 e^{j2\pi n} = e^0 + e^{j2\pi \cdot 1} + e^{j2\pi \cdot 2} \\ &= 1 + (\cos 2\pi + j \sin 2\pi) + (\cos 4\pi + j \sin 4\pi) \\ &= 1 + 1 + 1 = 3 = N \end{aligned}$$

Let $k=2$, $N=3$:

$$\sum_{n=0}^2 e^{j \frac{2\pi \cdot 2n}{3}} = \sum_{n=0}^2 e^{j \frac{2\pi \cdot 2n}{3}} = e^0 + e^{j \frac{2\pi \cdot 2(1)}{3}} + e^{j \frac{2\pi \cdot 2(2)}{3}}$$

$$= 1 + \left(\cos \frac{4\pi}{3} + j \sin \frac{4\pi}{3} \right) + \left(\cos \frac{8\pi}{3} + j \sin \frac{8\pi}{3} \right)$$

$$= 1 + (-0.5 + (-j0.866)) + (-0.5 + (+j0.866))$$

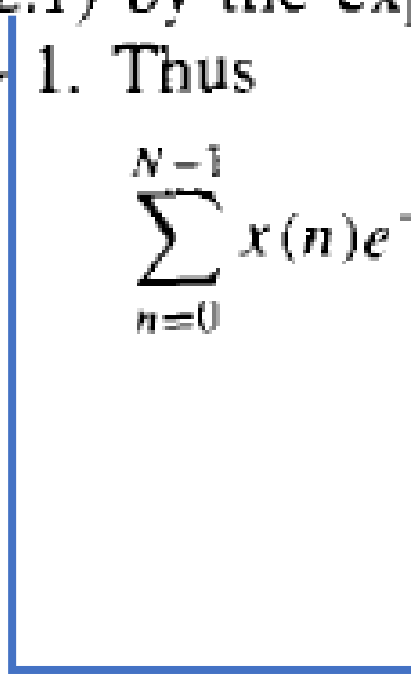
$$= 1 - 1 + 0 = 0$$

To derive the expression for the Fourier coefficients, we use the following formula:

$$\sum_{n=0}^{N-1} e^{j2\pi kn/N} = \begin{cases} N, & k = 0, \pm N, \pm 2N, \dots \\ 0, & \text{otherwise} \end{cases} \quad (4.2.2)$$

The expression for the Fourier coefficients c_k can be obtained by multiplying both sides of (4.2.1) by the exponential $e^{-j2\pi ln/N}$ and summing the product from $n = 0$ to $n = N - 1$. Thus

$$\sum_{n=0}^{N-1} x(n) e^{-j2\pi ln/N} = \sum_{n=0}^{N-1} \sum_{k=0}^{N-1} c_k e^{j2\pi(k-l)n/N} \quad (4.2.4)$$


$$x(n) = \sum_{k=0}^{N-1} c_k e^{j2\pi kn/N} \quad (4.2.1)$$

The expression for the Fourier coefficients c_k can be obtained by multiplying both sides of (4.2.1) by the exponential $e^{-j2\pi ln/N}$ and summing the product from $n = 0$ to $n = N - 1$. Thus

$$\sum_{n=0}^{N-1} x(n) e^{-j2\pi ln/N} = \sum_{n=0}^{N-1} \sum_{k=0}^{N-1} c_k e^{j2\pi(k-l)n/N} \quad (4.2.4)$$

If we perform the summation over n first, in the right-hand side of (4.2.4), we obtain

$$\sum_{n=0}^{N-1} e^{j2\pi(k-l)n/N} = \begin{cases} N & \text{when } k=l \\ 0 & \text{when } k \neq l \end{cases} \quad (4.2.5)$$

Let's verify each case.

$$\sum_{n=0}^{N-1} e^{j2\pi(n-l)n/N} = \begin{cases} N & \text{when } k=l \\ 0 & \text{when } k \neq l \end{cases}$$

Let's verify each case:

① $k=l$

$$\sum_{n=0}^{N-1} e^{j2\pi(0)n/N} = N$$

② $k \neq l$

$$\sum_n e^{j2\pi(n-l)n/N}$$

Let $e^{j2\pi(n-l)/N} = \alpha \Rightarrow \alpha^n$

By geometric series we know

$$\sum_{n=0}^{N-1} a^n = \frac{1-a^N}{1-a}$$

$$= \frac{1-a^N}{1-a}$$

$$= \frac{1 - e^{j2\pi \frac{(N-1)}{N}}}{1 - e^{j2\pi \frac{(N-1)}{N}}}$$

$$= \frac{1 - e^{j2\pi (N-1)}}{1 - e^{j2\pi \frac{(N-1)}{N}}}$$

As $e^{j2\pi (N-1)} = \underbrace{\cos(2\pi(N-1))}_{1} + j \underbrace{\sin(2\pi(N-1))}_0$

$$= \frac{1 - \{1 - 0\}}{1 - e^{j2\pi \frac{(N-1)}{N}}} = 0$$

"partial summation formula" shown below:

$$\sum_{n=0}^{N-1} \alpha^n = \sum_{n=0}^{\infty} \alpha^n - \sum_{n=N}^{\infty} \alpha^n = \frac{1}{1-\alpha} - \frac{\alpha^N}{1-\alpha} = \frac{1-\alpha^N}{1-\alpha}$$

Therefore, the right-hand side of (4.2.4)

reduces to Nc_l and hence

$$c_l = \frac{1}{N} \sum_{n=0}^{N-1} x(n) e^{-j2\pi ln/N} \quad l = 0, 1, \dots, N-1 \quad (4.2.6)$$

$$\sum_{n=0}^{N-1} x(n) e^{-j2\pi ln/N} = \sum_{n=0}^{N-1} \sum_{k=0}^{N-1} c_k e^{j2\pi(k-l)n/N} \quad (4.2.4)$$

Thus we have the desired expression for the Fourier coefficients in terms of the signal $x(n)$.

The relationships (4.2.1) and (4.2.6) for the frequency analysis of discrete-time signals are summarized below.

FREQUENCY ANALYSIS OF DISCRETE-TIME PERIODIC SIGNALS

Synthesis equation	$x(n) = \sum_{k=0}^{N-1} c_k e^{j2\pi kn/N} \quad (4.2.7)$
Analysis equation	$c_k = \frac{1}{N} \sum_{n=0}^{N-1} x(n) e^{-j2\pi kn/N} \quad (4.2.8)$

Equation (4.2.7) is often called the *discrete-time Fourier series* (DTFS).